

DHARAVATH NAGARAJU

Embedded Systems Engineer | Embedded AI & IoT Developer

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SUMMARY

Embedded Systems Engineer with hands-on experience building real hardware-software systems: CNN inference deployed on-device (Raspberry Pi 5, Arduino Nano 33 BLE), custom UART communication protocols, and sensor integration over I2C/SPI/BLE. Comfortable across the stack — firmware, signal acquisition, and edge AI deployment — with a background in electrical distribution systems (33/11kV) that grounds the hardware side of the work.

SKILLS

Embedded Systems: Arduino, Raspberry Pi, C/C++, firmware development, microcontroller programming, real-time systems, IoT, sensor integration

Embedded AI / Computer Vision: TensorFlow Lite, CNN inference, MediaPipe, OpenCV, edge AI deployment, gesture recognition

Communication Protocols: I2C, SPI, UART (custom hex packet protocols), BLE, PWM

Electrical Engineering: 33/11 kV distribution, transformers, relay protection, substation maintenance

Programming & Tools: Python, PySerial, Flask, Linux, Arduino IDE, PlatformIO, VS Code, MATLAB, Git, MS Office

Web & Dashboards: HTML, CSS, JavaScript, real-time data visualization

PROJECTS

AI-Based Hand Gesture Recognition — CNN, MediaPipe, Raspberry Pi 5, UART

- Delivered real-time gesture classification from a live camera feed by training a CNN on MediaPipe hand-landmark data and deploying inference on a Raspberry Pi 5
- Enabled real-time hardware control via a custom hexadecimal UART packet protocol (start byte, command, XOR checksum) built for direct embedded-system integration — github.com/nagaraju072001-design/Gesture-Recognition-CNN

Magic Wand — Edge AI Gesture Recognition Device

- Achieved fully on-device inference with zero cloud dependency by running a TensorFlow Lite CNN on an Arduino Nano 33 BLE Sense to classify IMU data in real time over I2C, with BLE output and RGB LED feedback

Smart Home Monitoring System + Sensor Dashboard

- Built an end-to-end IoT pipeline from raw sensor signal to browser display: temperature, humidity, and dust sensors wired over I2C/SPI with continuous UART data acquisition, feeding a responsive live-refresh web dashboard (HTML/CSS/JS)

Arduino RADAR & Stepper Motor Control

- Reached ~95% object detection accuracy (2–100cm range) with sub-1-second response time using ultrasonic sensing and PWM control
- Cut speed fluctuation 20–25% and held 1.8° step accuracy through optimized PWM-based stepper motor control logic

UPS Battery Monitoring with Automatic Changeover

- Reduced power interruption time by ~90% by designing relay-based automatic switching with real-time voltage monitoring

PROFESSIONAL EXPERIENCE

Process Associate | Genpact

Jul 2024 – Aug 2025

- Cut operational errors ~15% by introducing systematic double-verification checks across 250+ daily records spanning 10+ SOPs
- Sustained 98–99% processing accuracy at 250+ records/day while holding 100% compliance across quality, security, and reporting audits

Intern – Electrical Field Engineer | SPDCL, Telangana

Jun 2023 – Aug 2023

- Supported field operations across 33kV/11kV distribution systems and 10+ transformer maintenance cycles, applying strict electrical safety protocols throughout

Intern – Electrical Systems | AVR Power Suppliers, Hyderabad

Jun 2020 – Dec 2020

- Built foundational expertise in transformer design (step-up, step-down, isolation) and relay protection systems (overload, earth fault, differential) across the full power-flow chain

EDUCATION

M.S., Embedded Systems — Paris Institute of Digital Technology, France 2025–2027

B.Tech, Electrical & Electronics Engineering — JNTUH • CGPA: 7.8 2021–2024

Diploma, Electrical & Electronics Engineering — SBTET • CGPA: 8.45 2018–2021

LANGUAGES

English (C2) French (A2)